Tusk Robots E10 (Titan) Pallet AMR - Master CAD Design & Dimensioning Guide

This engineering guide details the exact mechanical dimensions, specifications, CAD blueprints, and component callouts for the **Tusk Robots E10 (Titan)** heavy-load low-profile pallet AMR (Invisible Fork Arm / Recessed Channel design).

1. Product Overview & Visual Identification

The **Tusk E10 (Titan)** is Tusk Robots' flagship high-capacity autonomous mobile robot designed for handling heavy pallets (up to 1,500 kg) in narrow aisles without traditional bulky fork masts.

Tusk Robots E10 (Titan) Low-Profile Pallet AMR (Realistic 3D CAD Render)

Key Architectural Characteristics (As Per E10 Titan Specs)

- **Body Configuration:** Low-profile rectangular chassis featuring silver-white side panels and a dark grey central bed with flush embedded fork channels.
- **Perimeter Safety & Status Lighting:** Green LED perimeter light strips running continuously around the chassis corners.
- **Obstacle Avoidance:** Dual 360° safety laser LiDAR sensors mounted on opposite corners inside protective yellow housings.
- **Side Panel Layout:** Emergency Stop button, battery charging port, and side brand markings (`30001 TUSKROBOTS`).

2. Master CAD Dimensioning Table (E10 Titan Specs)

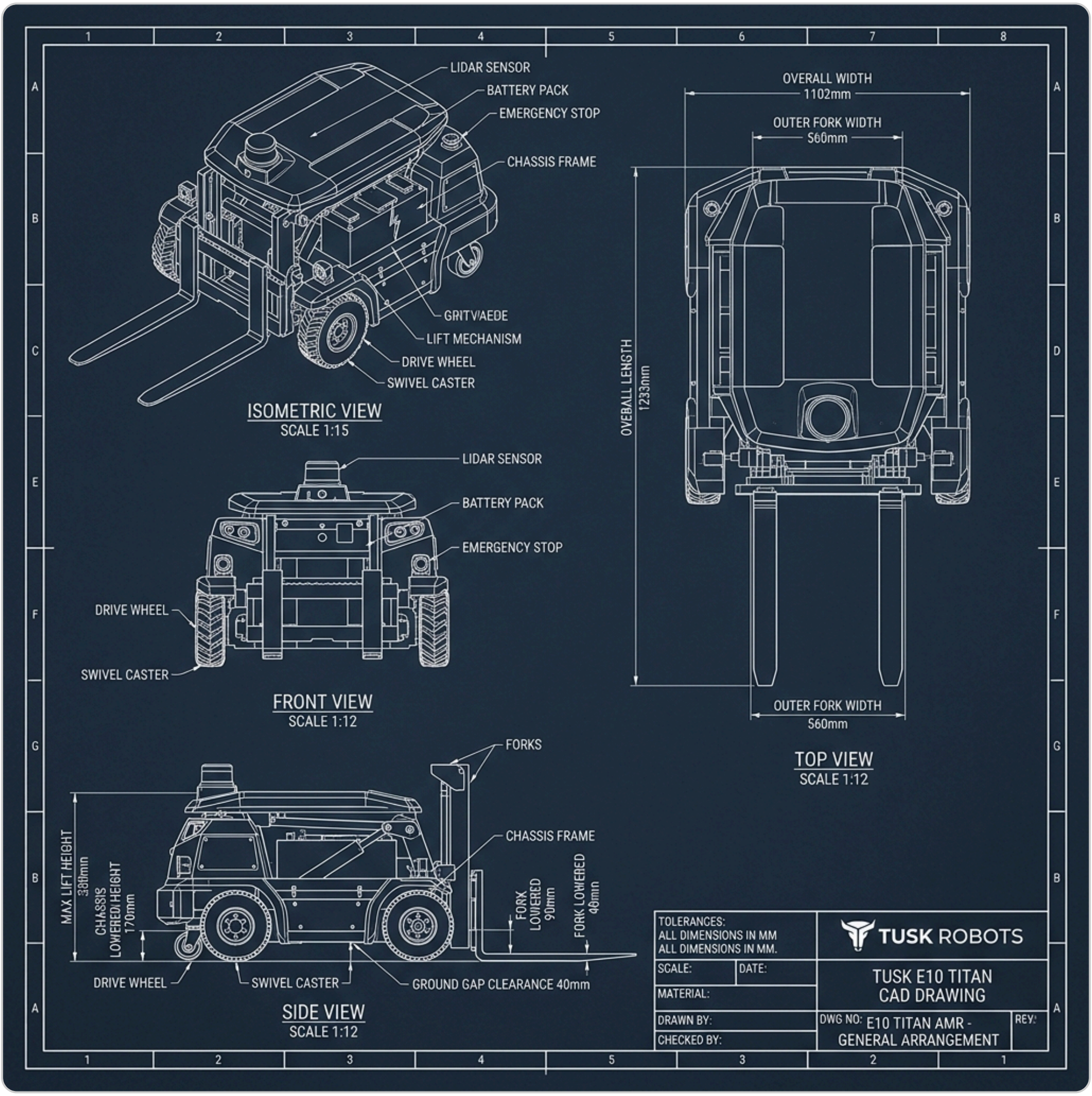
Use these exact dimensions when setting up sketch origins, planes, and extrusions in SolidWorks, Fusion 360, or Inventor:

Parameter / Feature	E10 (Titan) Dimension	Tolerance	CAD Modeling Notes
Rated Load Capacity	**1,200 kg** (Max **1,500 kg**)	—	Industrial heavy-payload class
Robot Self-Weight	**310 kg**	`\pm 5.0\text{ kg}`	Includes battery and drive motors
Overall Robot Length	**1233 mm** (Standard) / **1399 mm** (SLAM)	`\pm 1.0\text{ mm}`	Length of outer chassis envelope
Overall Robot Width	**1102 mm**	`\pm 1.0\text{ mm}`	Chassis outer side panel width
Chassis Lowered Height	**170 mm**	`\pm 0.5\text{ mm}`	Total height from floor to top deck
Fork Lowered Height	**90 mm**	`\pm 0.5\text{ mm}`	Has `7\text{ mm}` downward floating stroke
Max Lifting Height	**`\le 330\text{ mm}`**	`\pm 1.0\text{ mm}`	Total vertical lift travel: `160\text{ mm}`

Parameter / Feature	E10 (Titan) Dimension	Tolerance	CAD Modeling Notes
Fork Stretch / Reach Limit	**1400 mm max**	`\pm 2.0\text{ mm}`	Telescopic extension option
Outer Fork Assembly Width	**560 mm**	`\pm 0.5\text{ mm}`	Fits standard EUR-2 / ISO pallets
Inner Fork Gap	**240 mm**	`\pm 0.5\text{ mm}`	Width between left & right tynes
Individual Tyne Width	**160 mm**	`\pm 0.5\text{ mm}`	`\frac{560 - 240}{2} = 160\text{ mm}`
Ground Gap Clearance	**40 mm**	`\pm 0.5\text{ mm}`	Chassis bottom plate to floor
Step Crossing Capacity	**15 mm**	`\pm 1.0\text{ mm}`	Max obstacle step height
Max Slope / Gradeability	`4^\circ` **(** `7\%` **)**	—	Fully loaded gradeability limit
Max Travel Speed	**2.0 m/s** (Empty) / **1.5 m/s** (Loaded)	—	Closed-loop speed control

3. Master 2D CAD Blueprint Drawing Sheet (E10 Titan)

Use this engineering blueprint sheet for orthographic projection alignment and dimension placement:



Tusk Robots E10 Titan AMR 2D CAD Blueprint Drawing Sheet (Illustrative Schematic / Non-Realistic)

4. Component Bill of Materials (BOM) & Specifications

Subsystem	Component Name	Material / Spec	Quantity	Engineering Function
Chassis Frame	Base Weldment	Q235 Steel Tube (`80\times 60\times 4\text{ mm}`)	1 Set	Main structural load-bearing frame
Top Deck	Cover Plate	Q235 Steel Plate (`8\text{ mm}`)	1 Set	Recessed center deck with embedded tynes
Drive Motors	BLDC Gearmotors	750W 24V BLDC (1:20 Planetary Gearbox)	2 Pcs	Differential center-axis drive unit
Drive Wheels	Motorized Wheels	Polyurethane (`\varnothing 200\text{ mm} \times 65\text{ mm}` , `\varnothing 25\text{ mm}` Bore)	2 Pcs	High friction floor traction
Load Rollers	Tandem Rollers	Polyurethane 95A (`\varnothing 60\text{ mm} \times 180\text{ mm}`)	4 Pcs	Housed inside `198\text{ mm}` bogie housing
Lift Drive	Ball Screw	SFU2505 (Dia `25\text{ mm}` , Lead `5\text{ mm}` , `600\text{ mm}` L)	1 Pc	Driven by 1000W BLDC lift motor
Guide Rails	Linear Guides	HGR25 Rails (`1100\text{ mm}`) + `4\times` HGH25CA	2 Sets	Smooth vertical carriage elevation

Subsystem	Component Name	Material / Spec	Quantity	Engineering Function
Corner Casters	Swivel Casters	Heavy-Duty Low-Profile ($\varnothing 60\text{ mm}$ wheels)	4 Pcs	Supports 4 corners during tight turns
Safety Sensors	Laser LiDAR	360° Safety LiDAR Scanners	2 Pcs	Corner diagonal obstacle sensing
Battery	Battery Pack	24V / 48V LiFePO4 Battery Pack	1 Pc	$\ge 6\text{ hours}$ runtime, fast charging

5. Manufacturing & Tolerance Notes

1. **Machining Tolerances (ISO 2768-m):**

- Dimensions $0.5 - 6\text{ mm}$: $\pm 0.1\text{ mm}$
- Dimensions $6 - 30\text{ mm}$: $\pm 0.2\text{ mm}$
- Dimensions $30 - 120\text{ mm}$: $\pm 0.3\text{ mm}$
- Dimensions $120 - 400\text{ mm}$: $\pm 0.5\text{ mm}$
- Dimensions $400 - 1000\text{ mm}$: $\pm 0.8\text{ mm}$
- Dimensions $1000 - 2000\text{ mm}$: $\pm 1.0\text{ mm}$

2. **Welding Notes:**

- All structural frame joints require 6 mm continuous fillet MIG welds (ER70S-6 wire).

3. **Surface Treatment:**

- Exterior powder coating: Fine-textured Industrial Grey (RAL 7016) and Signal White (RAL 9003).